

Lecture 12:

Git and GitHub



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Python for Molecular Sciences

MSSE 272, 3 Units



GitHub

Outline

- **Motivation**

- **Git**

installation and VS Code

status, add and commit

compare differences

restoring from index, stage and commit

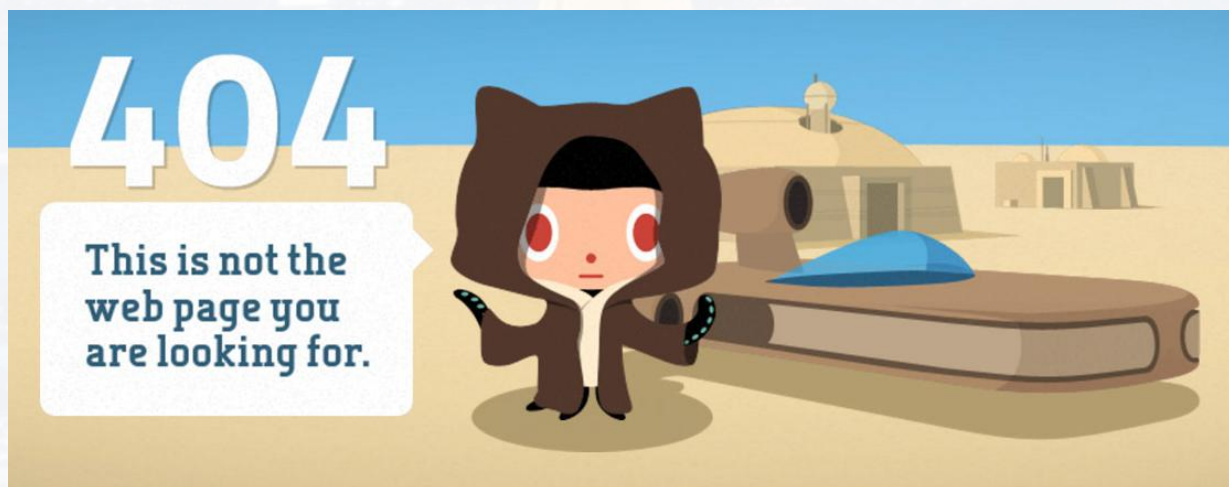
- **Git Hub**

create your repo

invite a co worker

push

fetch, merge, pull





GitHub

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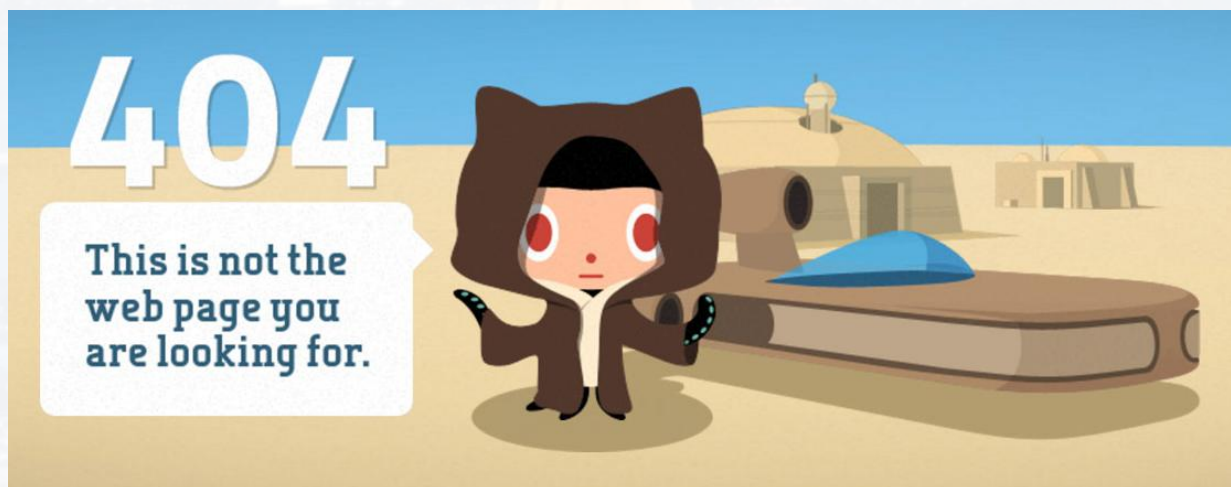
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When working on a project:

→ many different versions

```
#####  
def DistToPotential(x, y, a, b,  
    ...#takes locations of all part  
    .....  
    ...X.....= np.tile(x, (L, 1))  
    ...Y.....= np.tile(y, (L, 1))  
    .....  
    ...Dx.....= X - X.transpose()  
    ...Dy.....= Y - Y.transpose()
```

version 1

```
#####  
def DistToPotential(x, y, a, b,  
    ...#takes locations of all part  
    .....  
    ...X.....= np.tile(x, (L, 1))  
    ...Y.....= np.tile(y, (L, 1))  
    .....  
    ...Dx.....= X - X.transpose()  
    ...Dy.....= Y - Y.transpose()
```

version 1a

```
#####  
def DistToPotential(x, y, a, b,  
    ...#takes locations of all part  
    .....  
    ...X.....= np.tile(x, (L, 1))  
    ...Y.....= np.tile(y, (L, 1))  
    .....  
    ...Dx.....= X - X.transpose()  
    ...Dy.....= Y - Y.transpose()
```

version 2

```
#####  
def DistToPotential(x, y, a, b,  
    ...#takes locations of all part  
    .....  
    ...X.....= np.tile(x, (L, 1))  
    ...Y.....= np.tile(y, (L, 1))  
    .....  
    ...Dx.....= X - X.transpose()  
    ...Dy.....= Y - Y.transpose()
```

version 2a

```
#####  
def DistToPotential(x, y, a, b,  
    ...#takes locations of all part  
    .....  
    ...X.....= np.tile(x, (L, 1))  
    ...Y.....= np.tile(y, (L, 1))  
    .....  
    ...Dx.....= X - X.transpose()  
    ...Dy.....= Y - Y.transpose()
```

version 3

- easy to get **lost**
- What are the **differences between** the different **versions**?
- What if it turned out that **version 3c** was actually **better than 1f**?
- Sharing & exchanging with **co-workers**?



When working on a project:

→ many different versions

Version control system

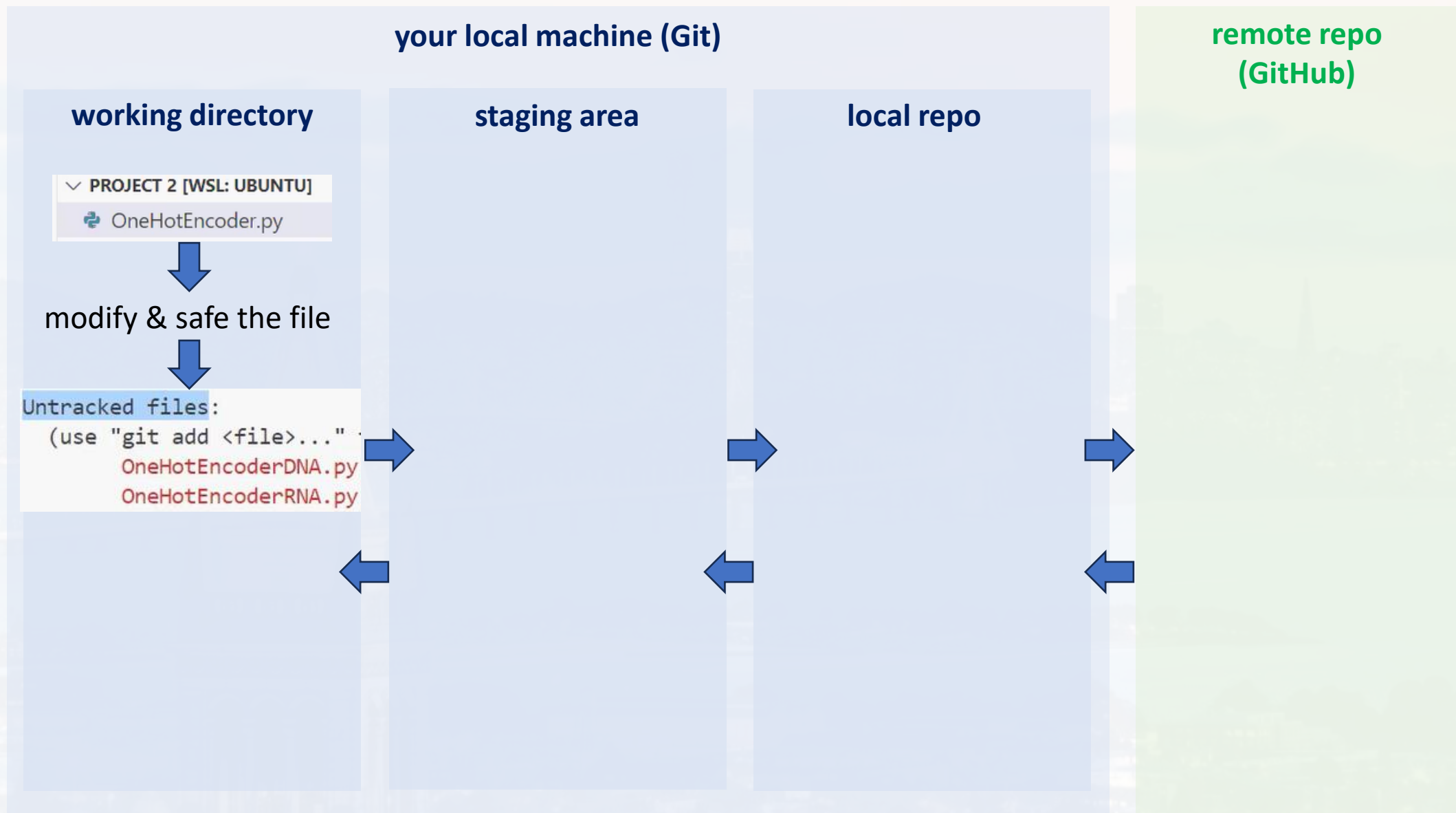
- keeps track of **different** versions
- keeps track of **changes**
- **coordinates** work with co-workers
- **stores** data efficiently



GitHub



structure





GitHub

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 - restoring from index, stage and commit

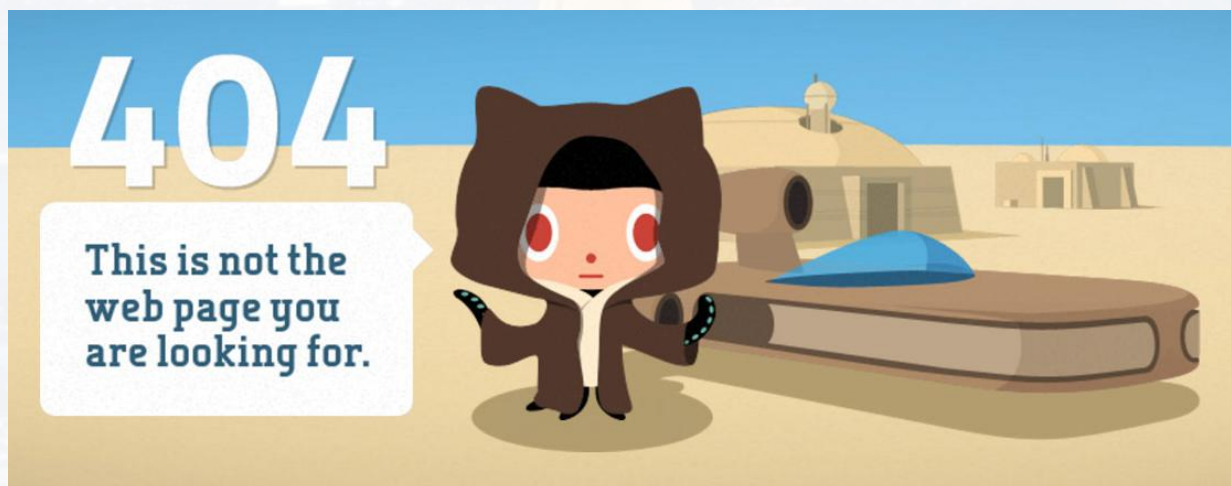
- **Git Hub**

 - create your repo

 - invite a co worker

 - push

 - fetch, merge, pull





1) open WSL terminal

note: Git should be installed already, nevertheless, let's run the following line:

2) install Git

```
mmh_user@DESKTOP-PPSA666:~$ sudo apt-get install git
```

3) confirm your password

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.34.1-1ubuntu1.12).
git set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 78 not upgraded.
```

4) check version:

```
(base) mmh_user@DESKTOP-PPSA666:~$ git --version
git version 2.34.1
```




4) check version:

```
(base) mmh_user@DESKTOP-PPSA666:~$ git --version  
git version 2.34.1
```

5) configure git: set your email address and user name

```
:~$ git config --global user.name "Markus Hohle"
```

```
:~$ git config --global user.email markus.hohle@berkeley.edu
```



4) check version:

```
(base) mmh_user@DESKTOP-PPSA666:~$ git --version  
git version 2.34.1
```

5) configure git: set your email address and user name

```
:~$ git config --global user.name "Markus Hohle"
```

```
:~$ git config --global user.email markus.hohle@berkeley.edu
```

Git has three levels of access:

- 1 **Local:** only for a **specific repo**
- 2 **Global:** across all repos for **currently logged-in user** (here: *mmh_user*)
- 3 **System:** **all users** and repos **on your machine**
(for example if you are admin or group leader etc)



4) check version:

```
(base) mmh_user@DESKTOP-PPSA666:~$ git --version  
git version 2.34.1
```

5) configure git: set your email address and user name

```
:~$ git config --global user.name "Markus Hohle"
```

```
:~$ git config --global user.email markus.hohle@berkeley.edu
```

6) check config: make sure everything is correct and type git config --list

```
(base) mmh_user@DESKTOP-PPSA666:~$ git config --list  
user.email=markus.hohle@berkeley.edu  
user.name=Markus Hohle  
init.defaultbranch=main
```



7) Git and VS Code:

create folder (say “git-lesson”) that you want to use as your git folder:

```
mkdir git-lesson
```

move to the folder:

```
cd git-lesson/
```

initialize your git folder by running `git init`

```
git init
```

open VS Code and you are all set

```
code .
```




7) Git and VS Code:

You might want to check the config again

```
(base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem280/git-lesson$ git config --list
user.email=markus.hohle@berkeley.edu
user.name=Markus Hohle
init.defaultbranch=main
credential.helper=/mnt/c/Program Files/Git/mingw64/bin/git-credential-manager-core.exe
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
(base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem280/git-lesson$
```

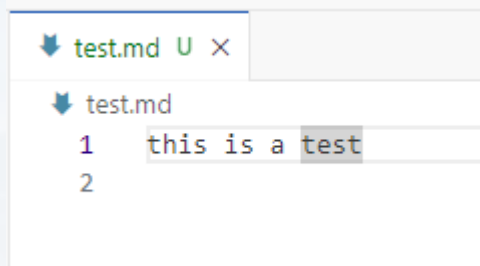
configs are all correct.
default branch is main (see later)



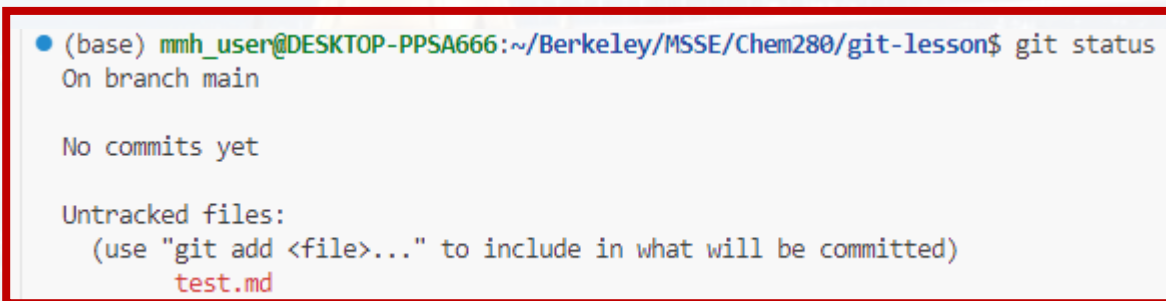
7) Git and VS Code:

Create a test file with any editor in your folder “git-lesson”

check if git recognizes the difference by running `git status`:



```
test.md U x
test.md
1  this is a test
2
```



```
(base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem280/git-lesson$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    test.md
```



If you see this, you are all set!



7) Git and VS Code: **Alternatively!**

- open VS code
- navigate to the folder which you want to use for your project

The screenshot shows the Visual Studio Code interface. On the left, the Explorer sidebar displays a project named 'PROJECT 2 [WSL: UBUNTU]' with a file 'OneHotEncoder.py'. A red box highlights the Git icon (a branching diagram) in the sidebar, with a red arrow pointing to it and the text 'click on the Git symbol'. On the right, the Welcome panel is visible, showing options like 'New File...', 'Open File...', 'Open Folder...', 'Clone Git Repository...', and 'Connect to...'. A red box highlights the 'Initialize Repository' button, with a red arrow pointing to it and the text 'click on Initialize Repository'. A tooltip on the right explains that the folder doesn't have a Git repository and provides instructions on how to initialize one and publish to GitHub.

click on the Git symbol

click on Initialize Repository

The folder currently open doesn't have a Git repository. You can initialize a repository which will enable source control features powered by Git.

Initialize Repository

To learn more about how to use Git and source control in VS Code [read our docs](#).

You can directly publish this folder to a GitHub repository. Once published, you'll have access to source control features powered by git and GitHub.

Publish to GitHub



7) Git and VS Code:

default branch name is
"main", which can be
renamed

watch:

12a_Installing_Git

12b_Git_and_VS_Code_in_WSL

The screenshot shows the VS Code interface with the Git source control panel on the left. A red box highlights the 'main*' branch in the source control panel. A red arrow points from the 'main*' branch to a dropdown menu that appears when clicking the branch name. The dropdown menu contains the following options:

- Select a branch or tag to checkout
- + Create new branch...
- + Create new branch from...
- Checkout detached...

The terminal window at the bottom shows the output of the 'git help' command, listing various Git commands and their descriptions:

```
branch      List, create, or delete branches
commit      Record changes to the repository
merge       Join two or more development histories together
rebase      Reapply commits on top of another base tip
reset       Reset current HEAD to the specified state
switch      Switch branches
tag         Create, list, delete or verify a tag object signed with GPG

collaborate (see also: git help workflows)
fetch       Download objects and refs from another repository
pull        Fetch from and integrate with another repository or a
push        Update remote refs along with associated objects

'git help -a' and 'git help -g' list available subcommands and some
concept guides. See 'git help <command>' or 'git help <concept>'
to read about a specific subcommand or concept.
See 'git help git' for an overview of the system.
(base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem272/Projects/Pi
```




GitHub

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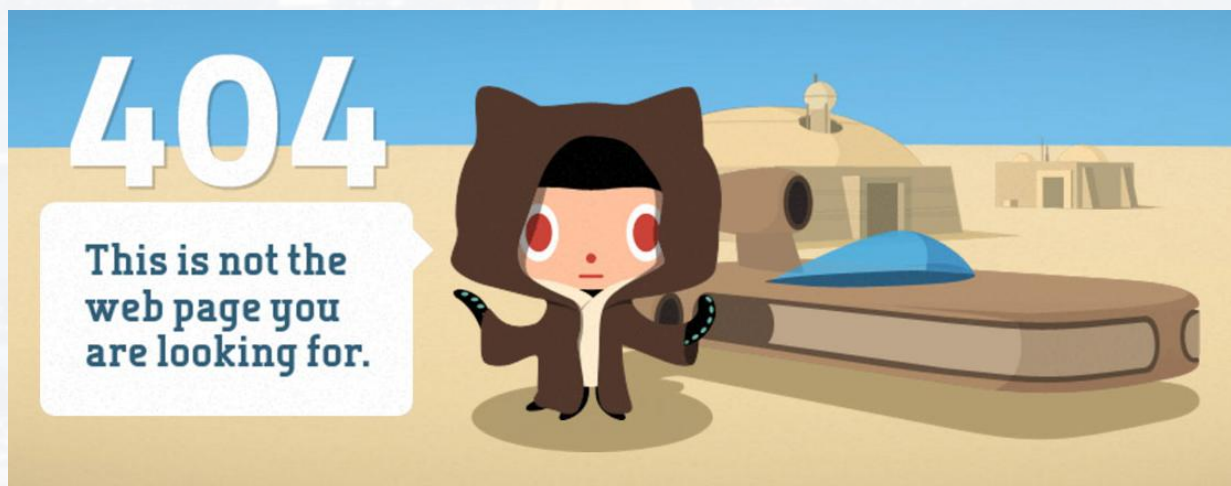
- **Git Hub**

create your repo

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fetch, merge, pull





structure



your local machine (Git)

working directory

PROJECT 2 [WSL: UBUNTU]

OneHotEncoder.py

modify & save the file

Git notices the change!



git status

Untracked files:

```
(use "git add <file>..."  
OneHotEncoderDNA.py  
OneHotEncoderRNA.py
```

staging area

local repo

remote repo (GitHub)

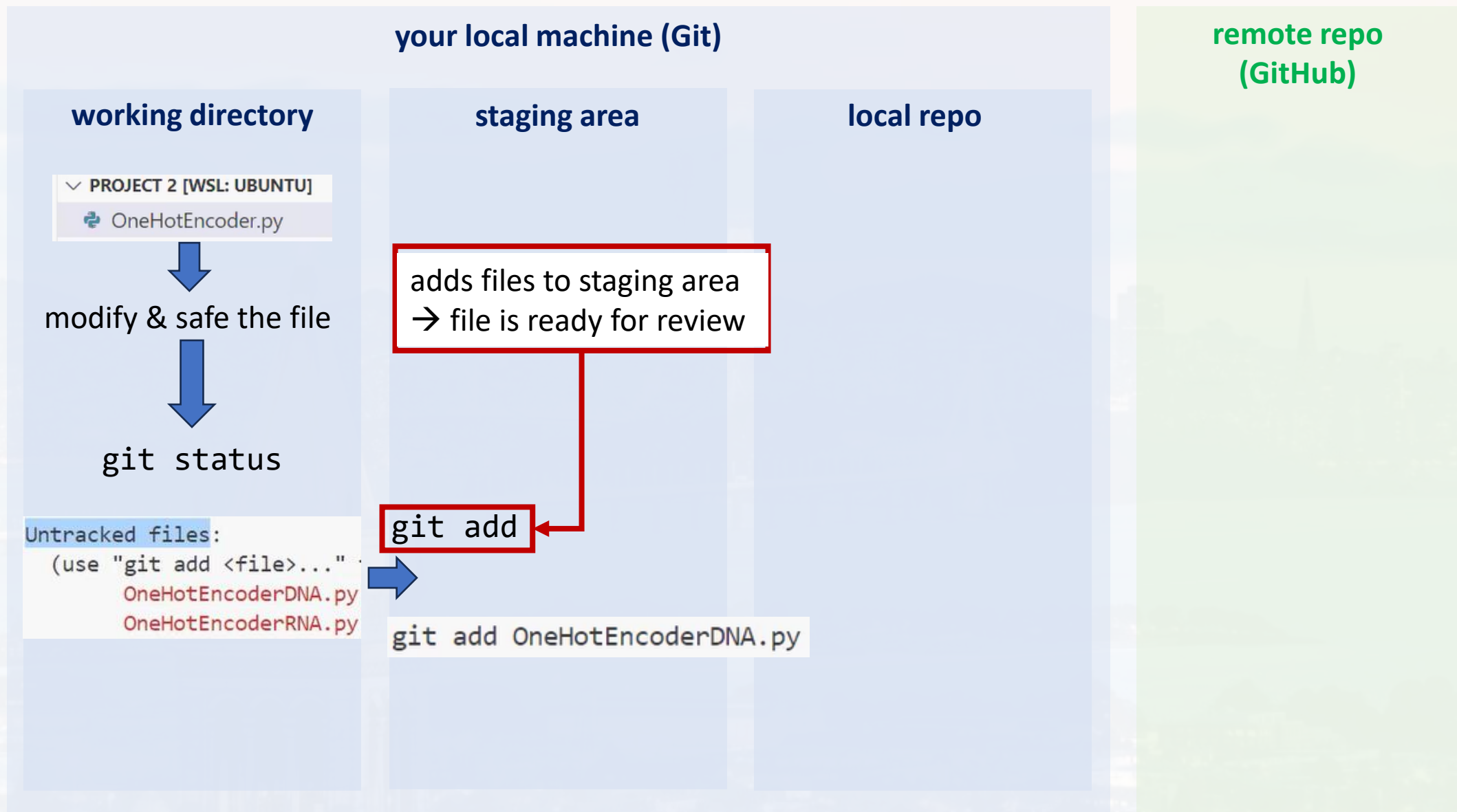
always check Git status

→ which files have been modified

→ which files are in the staging area

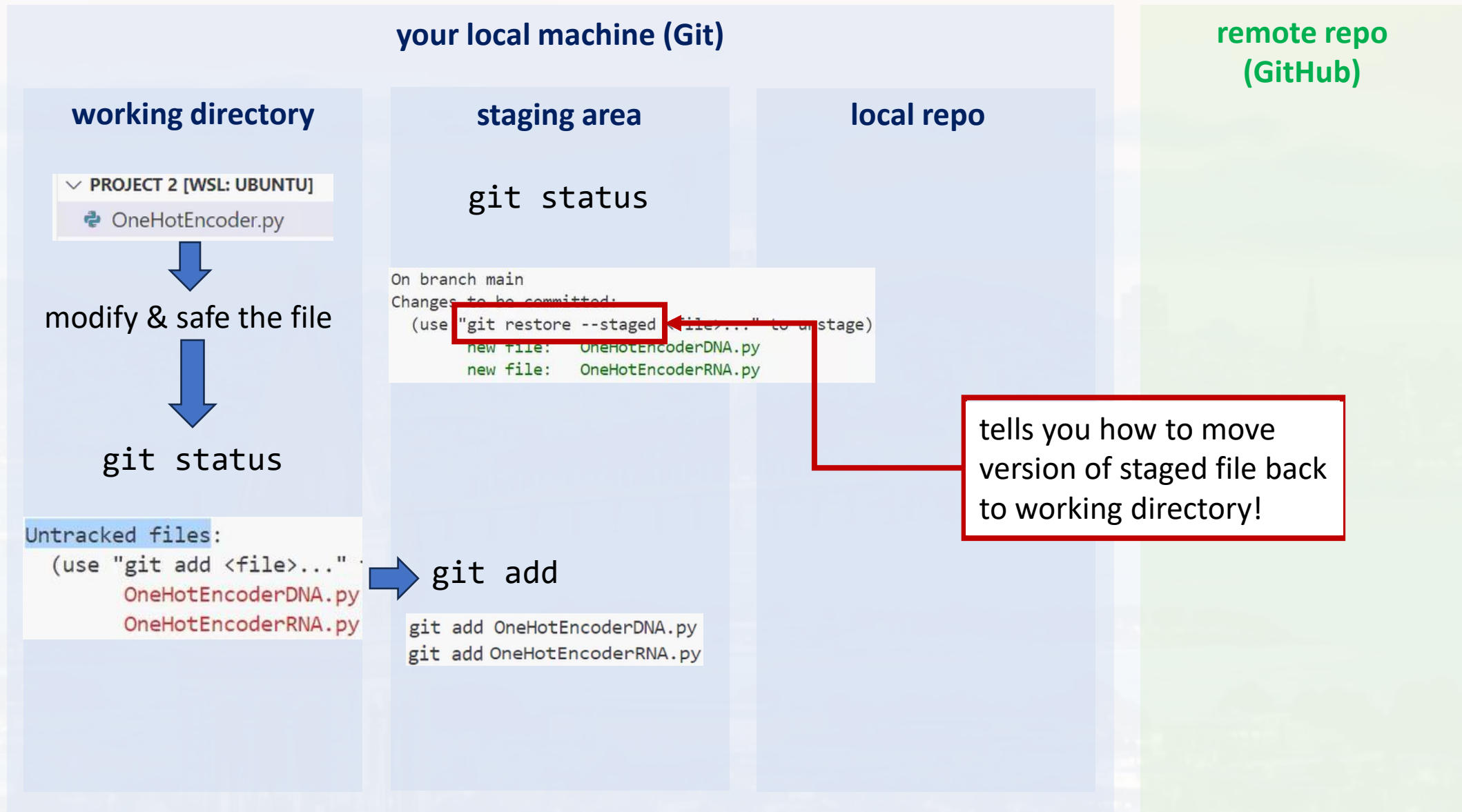


structure



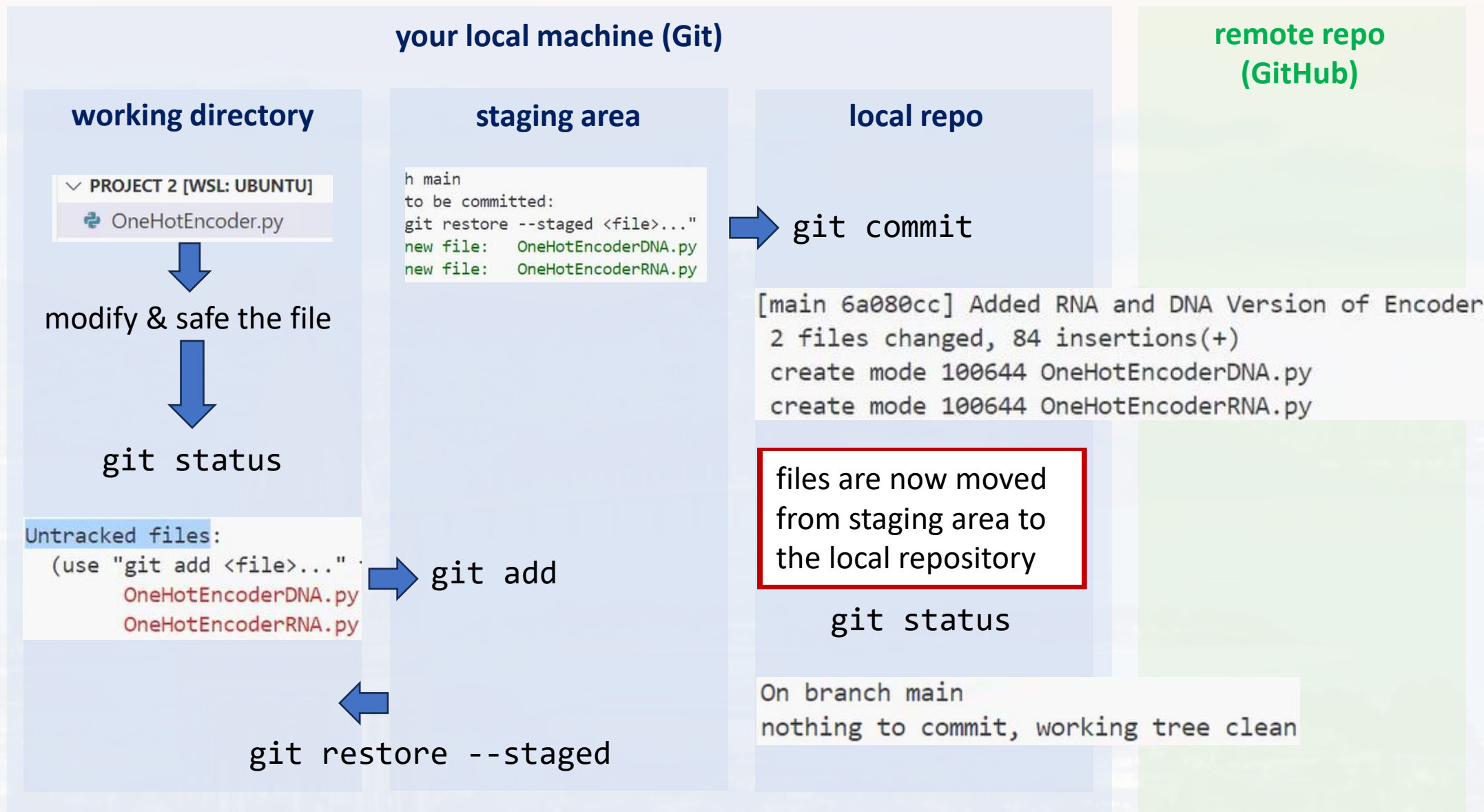


structure



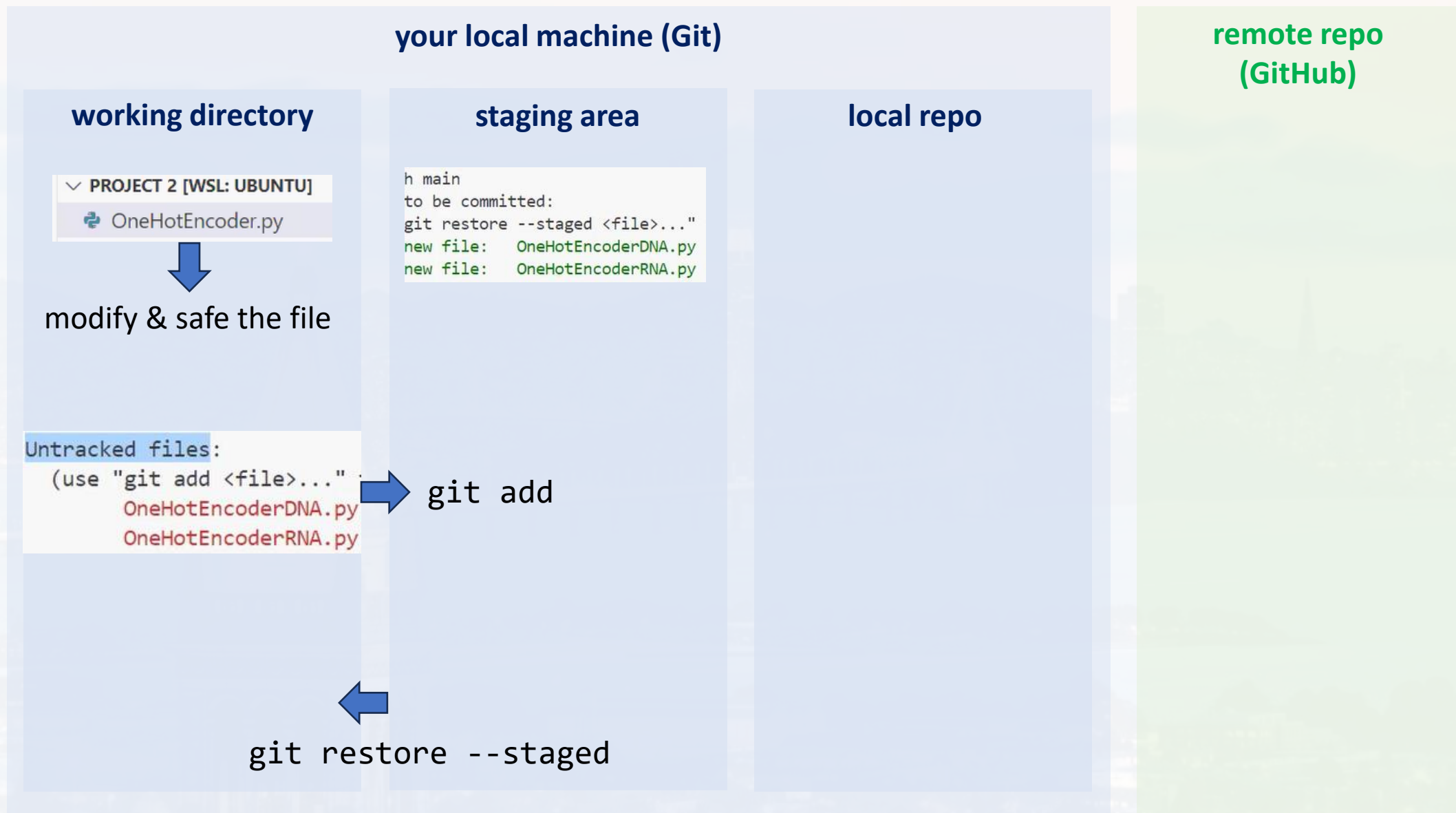


structure



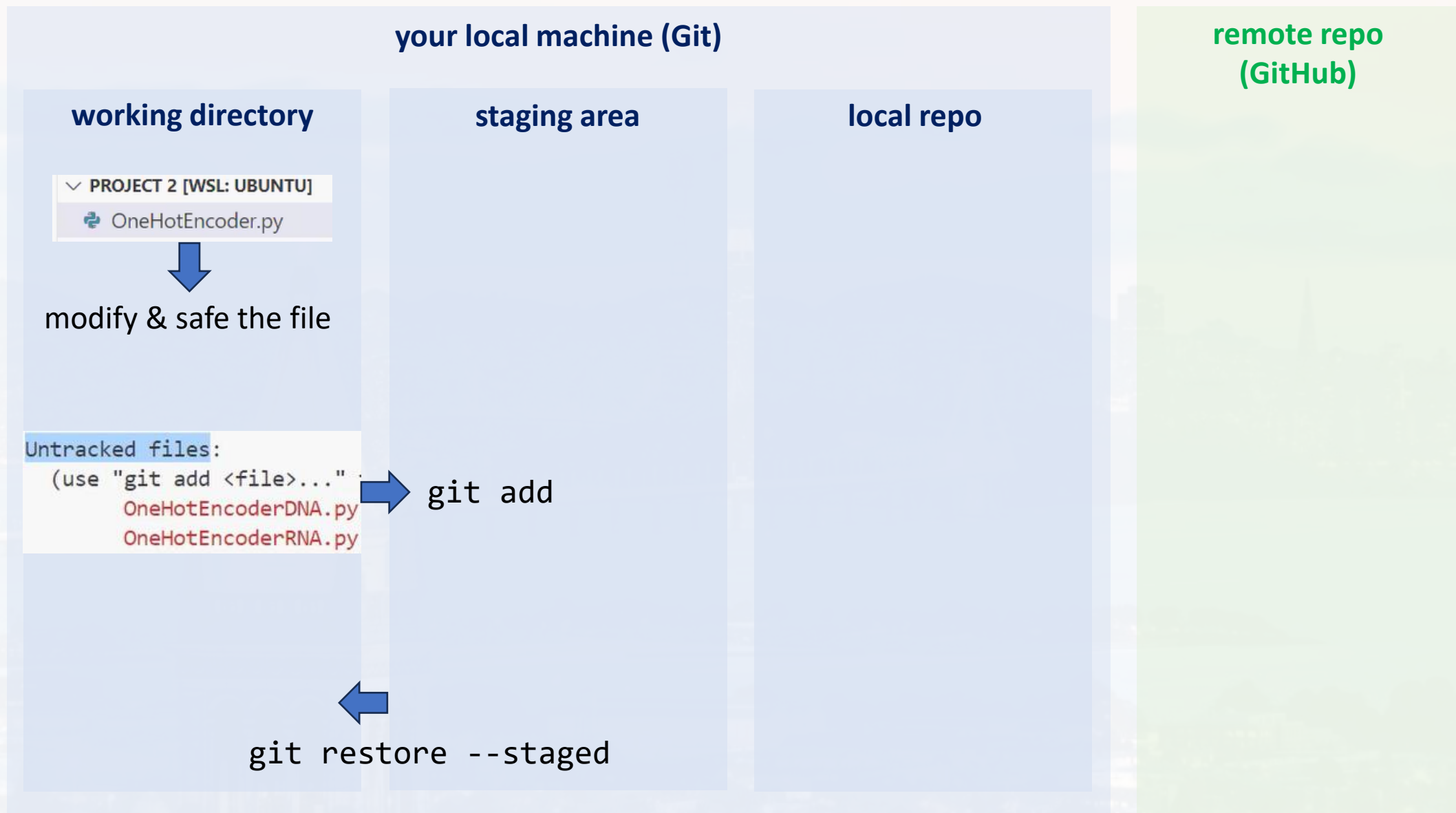


structure





structure





structure



your local machine (Git)

working directory

▼ PROJECT 2 [WSL: UBUNTU]
OneHotEncoder.py



modify & save the file

Untracked files:

(use "git add <file>..."
OneHotEncoderDNA.py
OneHotEncoderRNA.py



staging area

```
h main
to be committed:
git restore --staged <file>..."
new file:   OneHotEncoderDNA.py
new file:   OneHotEncoderRNA.py
```

git add

local repo

git commit

```
[main 6a080cc] Added RNA and DNA Version of Encoder
2 files changed, 84 insertions(+)
create mode 100644 OneHotEncoderDNA.py
create mode 100644 OneHotEncoderRNA.py
```

remote repo (GitHub)

watch:
12c_Git_commit



GitHub

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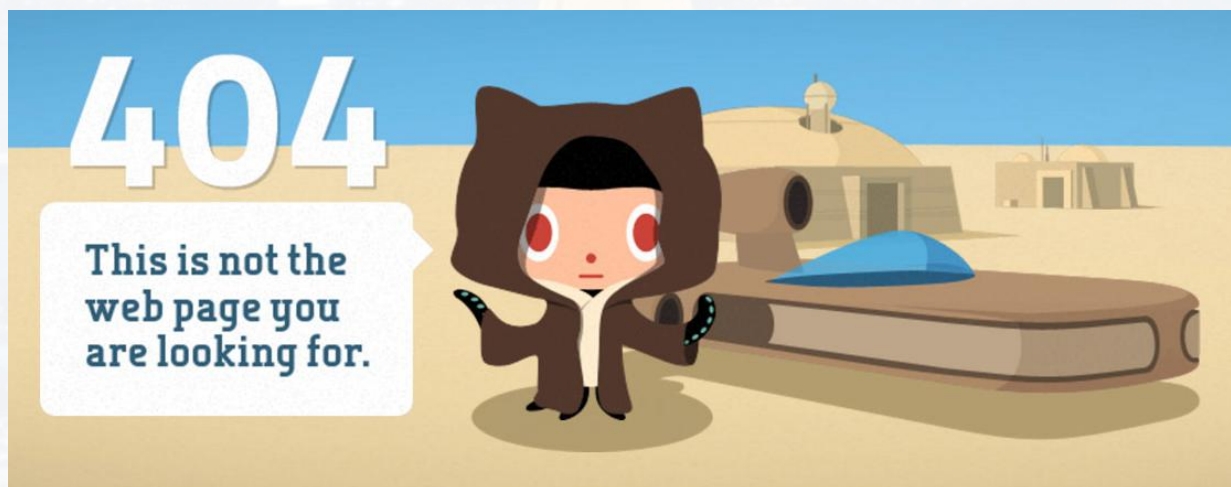
- Git Hub

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1) A new file (OneHotEncoder.py) in the working directory

`git status`

On branch main

Changes not staged for commit:

(use "git add <file>..." to update what will be committed)

(use "git restore <file>..." to discard changes in working directory)

modified: OneHotEncoder.py

`git add` `git add OneHotEncoder.py`

`git status`

On branch main

Changes to be committed:

(use "git restore --staged <file>..." to unstage)

modified: OneHotEncoder.py

2) We kept working on the file and have two versions now!



2) We kept working on the file and have two versions now!

git status

On branch main

Changes to be committed:

(use "git restore --staged <file>..." to unstage)

modified: OneHotEncoder.py

Changes not staged for commit:

(use "git add <file>..." to update what will be committed)

(use "git restore <file>..." to discard changes in working directory)

modified: OneHotEncoder.py

the **previous** version
which is located
at the **stage area**

the **current** version
which is located
at the **work folder**

Message (Ctrl+Enter to com...)

✓ Commit

✓ Staged Changes

1

OneHotEncoder.py

4, M

✓ Changes

1

OneHotEncoder.py

4, M

before adding or committing to the repo
→ review changes!



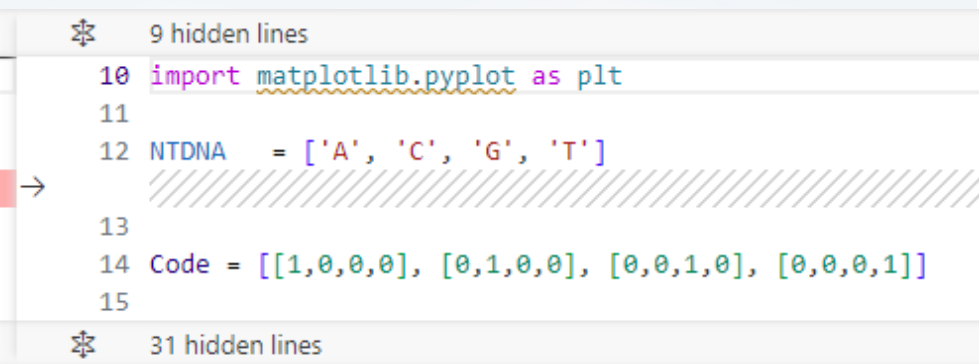
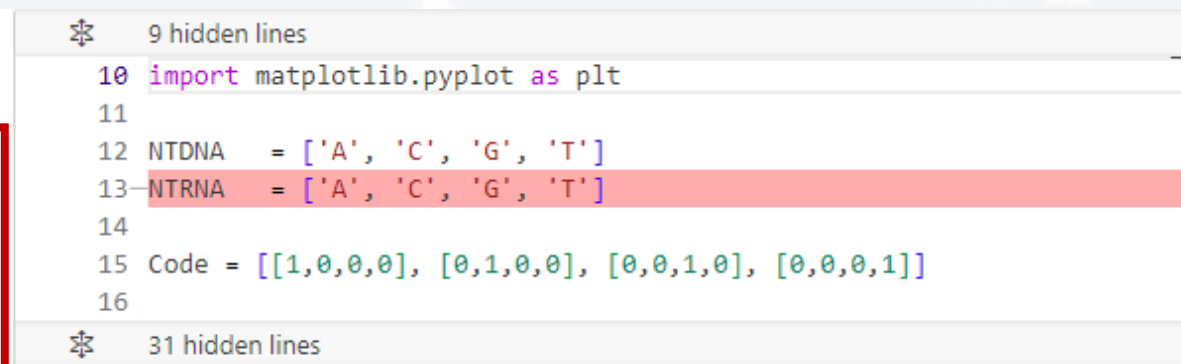
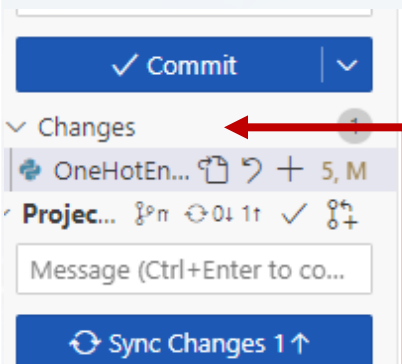
2) We kept working on the file and have two versions now!

before adding or committing to the repo

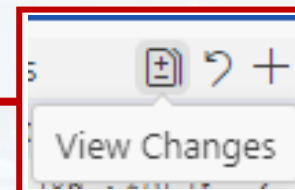
→ **review changes!**

staged version has one more line
(which got removed, hence **highlighted in red**)

folder version has one fewer line



click on *Changes* or *Staged Changes*



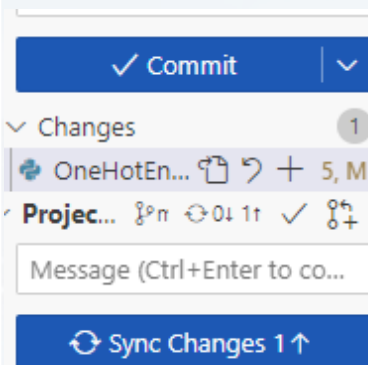


2) We kept working on the file and have two versions now!

before adding or committing to the repo

→ review changes!

staged version



```
9 hidden lines
10 import matplotlib.pyplot as plt
11
12 NTDNA = ['A', 'C', 'G', 'T']
13 NTRNA = ['A', 'C', 'G', 'T']
14
15 Code = [[1,0,0,0], [0,1,0,0], [0,0,1,0], [0,0,0,1]]
16
31 hidden lines
```

folder version has one more line
(hence **highlighted in green**)

```
9 hidden lines
10 import matplotlib.pyplot as plt
11
12 NTDNA = ['A', 'C', 'G', 'T']
13 NTRNA = ['A', 'C', 'G', 'T']
14+
15+ #an additional line
16
17 Code = [[1,0,0,0], [0,1,0,0], [0,0,1,0], [0,0,0,1]]
18
31 hidden lines
```

once we add or commit, Git tells us the changes:

```
[main b4d5c49] RNA and DNA Encoder added
1 file changed, 1 insertion(+), 1 deletion(-)
```

watch:

12d_Git_CompareDifferences



GitHub

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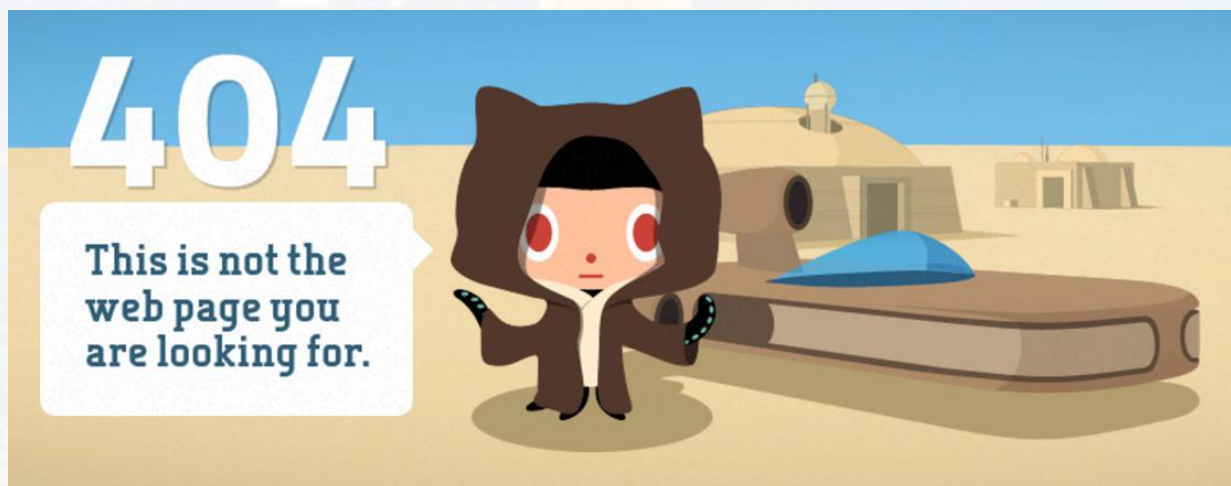
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We can always rescue any file from any state of our project!

- 1) changing file in working directory

```
undo: git restore
```

- 2) git add file from working directory to stage area

```
undo: git restore --staged
```

watch:

12e_Git_RestoreFromIndexAndStage

What if we have committed many different versions to the repo
but need to restore one particular version??



structure



your local machine (Git)

working directory

staging area

local repo

git log

```
commit b4d5c498d4d5db1d7a6cffa614ead30c1902aca2 (HEAD -> main)
Author: MarkusMSSE_Chem272 <markus.hohle@berkeley.edu>
Date: Thu Jul 10 23:51:55 2025 +0200
```

RNA and DNA Encoder added

```
commit 124f919fcfbfd72c3df0de1180c1e450d8dd2be5
Author: MarkusMSSE_Chem272 <markus.hohle@berkeley.edu>
Date: Thu Jul 10 23:45:21 2025 +0200
```

Test

```
commit 6a080ccee9b3ba8bde5151b810fcf0223e482f9e
Author: MarkusMSSE_Chem272 <markus.hohle@berkeley.edu>
Date: Thu Jul 10 22:16:51 2025 +0200
```

Added RNA and DNA Version of Encoder

```
commit 542391b0a769e142f4ef029c722abbecb3cdddf8
Author: MarkusMSSE_Chem272 <markus.hohle@berkeley.edu>
Date: Thu Jul 10 22:00:58 2025 +0200
```

message

remote repo
(GitHub)



The last commit (HEAD)

git log

```
commit b4d5c498d4d5db1d7a6cffa614ead30c1902aca2 (HEAD -> main)
Author: MarkusMSSE_Chem272 <markus.hohle@berkeley.edu>
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```

message



git log

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Date: Thu Jul 10 22:00:58 2025 +0200

message

each commit has an **individual ID**



git log

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```

message

each commit has a **date**



git log

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```

message

the **messages** you added are also stored



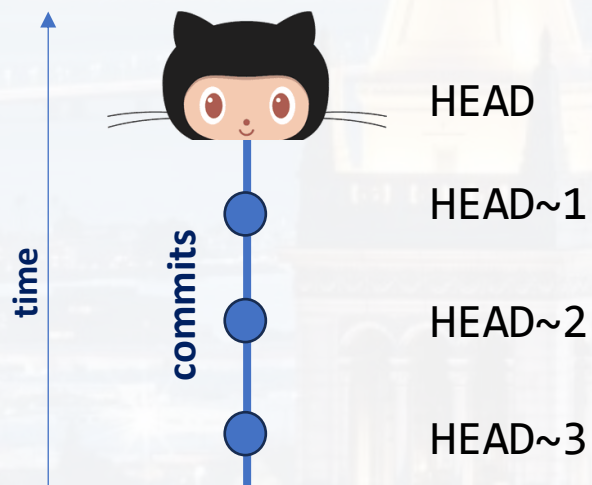
if we want to move a file from a given **commit back to the staging area**:

→ either referring to the **ID**

→ or to the **head**

```
git restore --source=HEAD~2 OneHotEncoder.py
```

two commits before the current one



git log

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Date: Thu Jul 10 22:16:51 2025 +0200
```

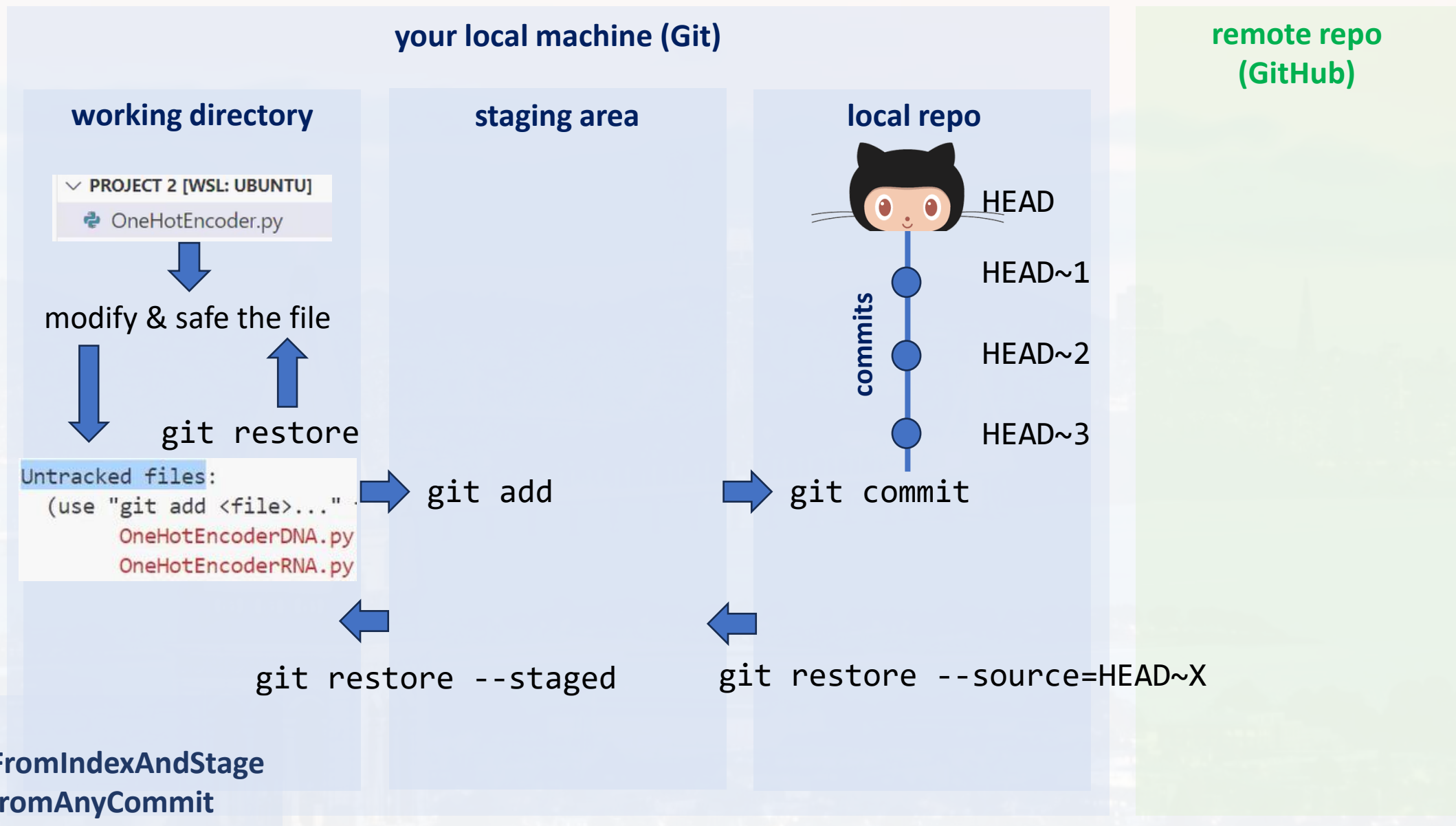
Added RNA and DNA Version of Encoder

```
commit 542391b0a769e142f4ef029c722abbeeb3cdddf8
Author: MarkusMSSE_Chem272 <markus.hohle@berkeley.edu>
Date: Thu Jul 10 22:00:58 2025 +0200
```

message



structure



watch:

12e_Git_RestoreFromIndexAndStage

12f_Git_RestoreFromAnyCommit



GitHub

Outline

- Motivation

- Git

installation and VS Code

status, add and commit

compare differences

restoring from index, stage and commit

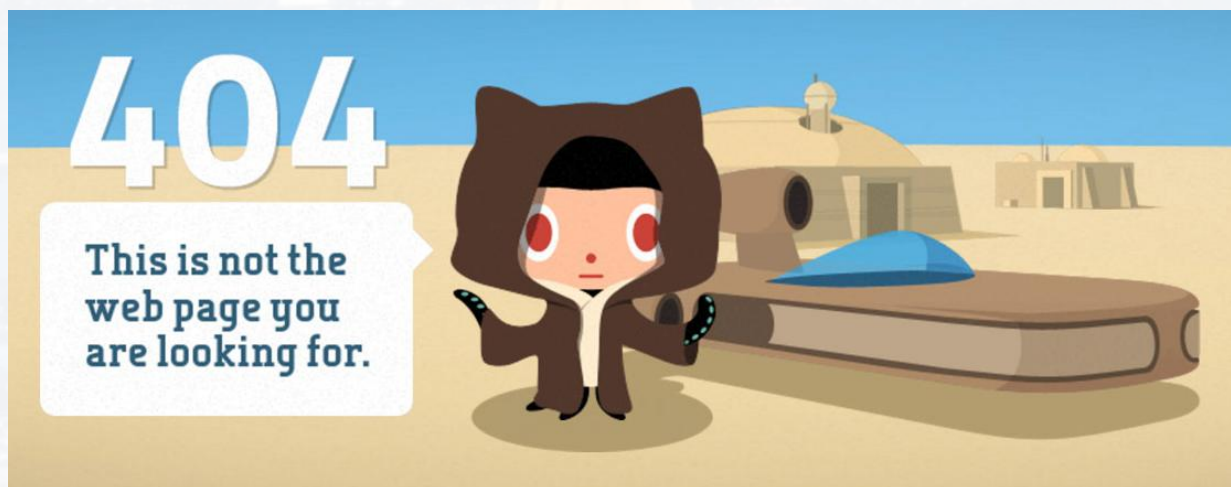
- **Git Hub**

create your repo

invite a co worker

push

fetch, merge, pull





structure



your local machine (Git)

working directory

staging area

local repo

**remote repo
(GitHub)**



got to github.com

Build and ship software on a single, collaborative platform

Join the world's most widely adopted AI-powered developer platform.

[Sign up for GitHub](#)[Try GitHub Copilot](#)

Sign up to GitHub

Email*

Password*

Password should be at least 15 characters OR at least 8 characters including a number and a lowercase letter.

Username*

Username may only contain alphanumeric characters or single hyphens, and cannot begin or end with a hyphen.

Your Country/Region*

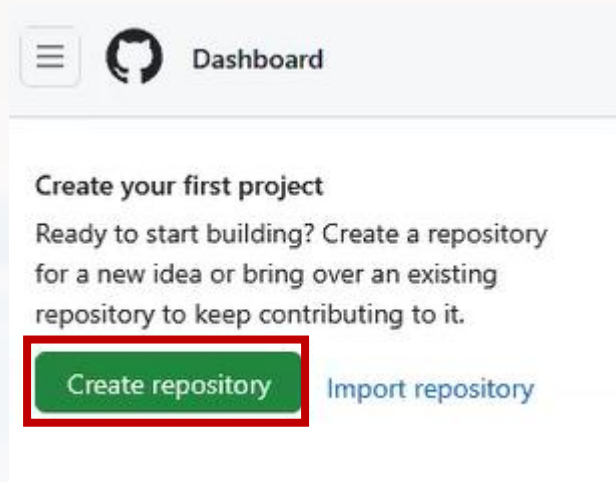
For compliance reasons, we're required to collect country information to send you occasional updates and announcements.

remote repo
(GitHub)

register and confirm email address



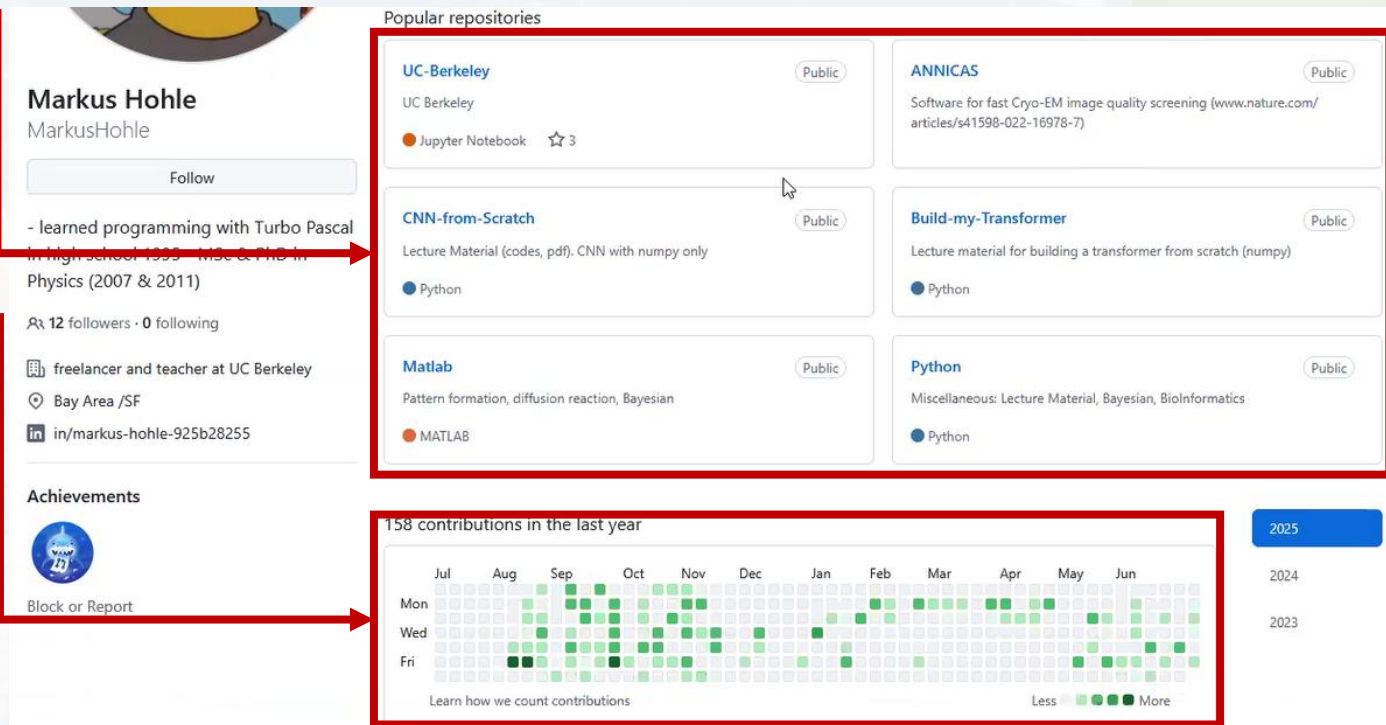
create new repository



remote repo
(GitHub)

list of repos

your **garden** shows
your activity:
keep it green!





create new repository



Create your first project

Ready to start building? Create a repository for a new idea or bring over an existing repository to keep contributing to it.

Create repository

[Import repository](#)

Create a new repository [Try the new experience](#)

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository.](#)

Required fields are marked with an asterisk (*).

Owner *



MarkusMSSEChem272

Repository name *

I

Great repository names are short and memorable. Need inspiration? How about [super-broccoli](#) ?

Description (optional)



Public

Anyone on the internet can see this repository. You choose who can commit.



Private

You choose who can see and commit to this repository.

Initialize this repository with:



Add a README file

This is where you can write a long description for your project. [Learn more about READMEs.](#)

Add .gitignore

.gitignore template: None



create new repository

add a meaningful name...

...and discription...

public means **anyone**
can see your repo

you can add more info
in a ReadMe file

gitignore lists files and
directories that will
not be committed if
you use the option -a

Create a new repository [Try the new experience](#)

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository.](#)

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This is where you can write a long description for your project. [Learn more about READMEs.](#)

Add .gitignore

.gitignore template: None



create new repository

Create repository

Congratulations! You have created your first repository!

The screenshot shows the GitHub interface for a newly created repository named "Project2", which is public. At the top, there are buttons for "Pin", "Watch" (0), "Fork" (0), and "Star" (0). Below this, the repository is shown with a single branch named "main" and 0 tags. A search bar "Go to file" and buttons for "Add file" and "Code" are visible. The commit history shows an "Initial commit" by "MarkusMSSEChem272" with the hash "0ae54c5" and the message "Initial commit". A file named "README.md" is listed as part of this commit. The README content is displayed below, featuring the title "Project2" and the subtitle "Project 2 for Chem 272". On the right side, the "About" section provides details: "Project 2 for Chem 272", "0 stars", "0 watching", and "0 forks". It also includes links for "Readme", "Activity", "Releases" (with a note "No releases published" and a link "Create a new release"), and "Packages" (with a note "No packages published" and a link "Publish your first package").

Project2 Public

main 1 Branch 0 Tags

Go to file Add file <> Code

MarkusMSSEChem272 Initial commit 0ae54c5 · now 1 Commit

README.md Initial commit now

README

Project2

Project 2 for Chem 272

About

Project 2 for Chem 272

- Readme
- Activity
- 0 stars
- 0 watching
- 0 forks

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)



Note there are way more settings you can adjust

[Actions](#) [Projects](#) [Wiki](#) [Security](#) [Insights](#) [Settings](#)

General

Access

Collaborators

Moderation options

Code and automation

Branches

Tags

Rules

Actions

Models

Webhooks

Copilot

Environments

Codespaces

Pages

General

Repository name

Project2 Rename

☐ **Template repository**
Template repositories let users generate new repositories with the same

☐ **Require contributors to sign off on web-based commits**
Enabling this setting will require contributors to sign off on commits made by contributors to affirm that their commit complies with the repository's [more about signing off on commits](#).

Default branch

The default branch is considered the "base" branch in your repository. It is automatically made, unless you specify a different branch.

main

Social preview



GitHub

Outline

- Motivation

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installation and VS Code

status, add and commit

compare differences

restoring from index, stage and commit

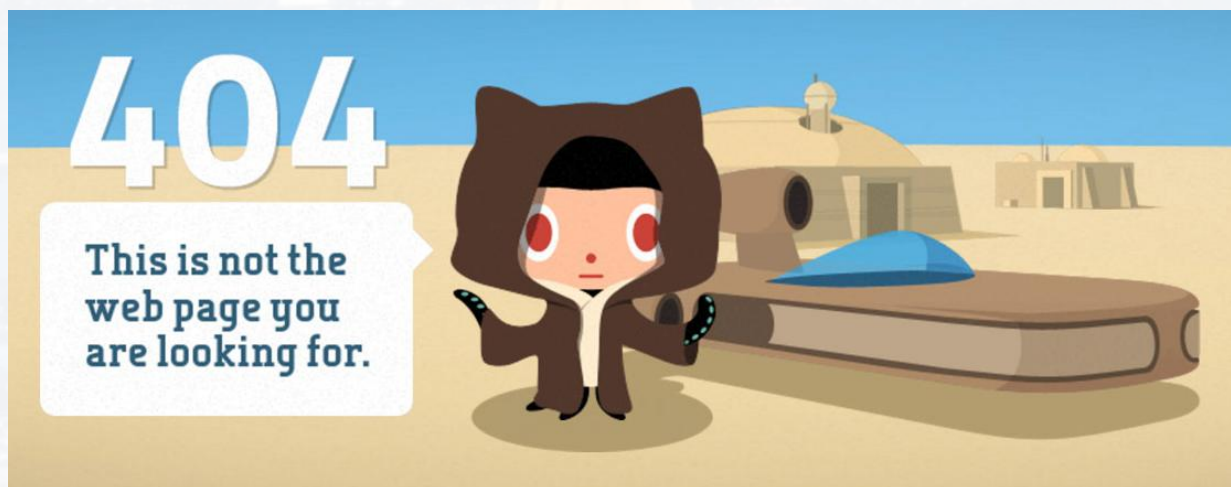
- **Git Hub**

create your repo

invite a co worker

push

fetch, merge, pull





go to **settings**

The screenshot shows the GitHub interface for a repository named 'Project2' owned by 'MarkusMSSEChem272'. The repository is currently on the 'main' branch, and the selected file is 'README.md'. A commit by 'MarkusMSSEChem272' titled 'Update README.md' is visible. The 'Settings' tab in the top navigation bar is highlighted with a red box and a hand cursor, indicating the next step in the tutorial. The background of the page features a faint image of the University of California Berkeley campus.

MarkusMSSEChem272 / Project2

<> Code Issues Pull requests Actions Projects Wiki Security Insights **Settings**

main Project2 / README.md

MarkusMSSEChem272 Update README.md

Preview Code Blame 3 lines (3 loc) · 61 Bytes Code 55% faster with GitHub Copilot

Project2



go to **collaborators**

⚙️ General

Access

👤 Collaborators

Moderation options

Code and automation

🌿 Branches

🏷️ Tags

📄 Rules

🎬 Actions

🔗 Models

🔗 Webhooks

🤖 Copilot

📁 Environments

📄 Codespaces

📄 Pages

General

Repository name

Project2

☐ Template repository

☐ Require contributors to sign off on their commits

Default branch

main

Add people to Project2

Search by username, full name, or email



Markus Hohle
MarkusHohle • Invite collaborator

Cancel

Add to repository

add **collaborators**



MarkusMSSEChem272 invited you to collaborate on
MarkusMSSEChem272/Project2

Accept invitation

Decline invitation

your **collaborator** will get an invitation and needs to confirm it

watch:
12h_GitHub_InviteCoWorker



GitHub

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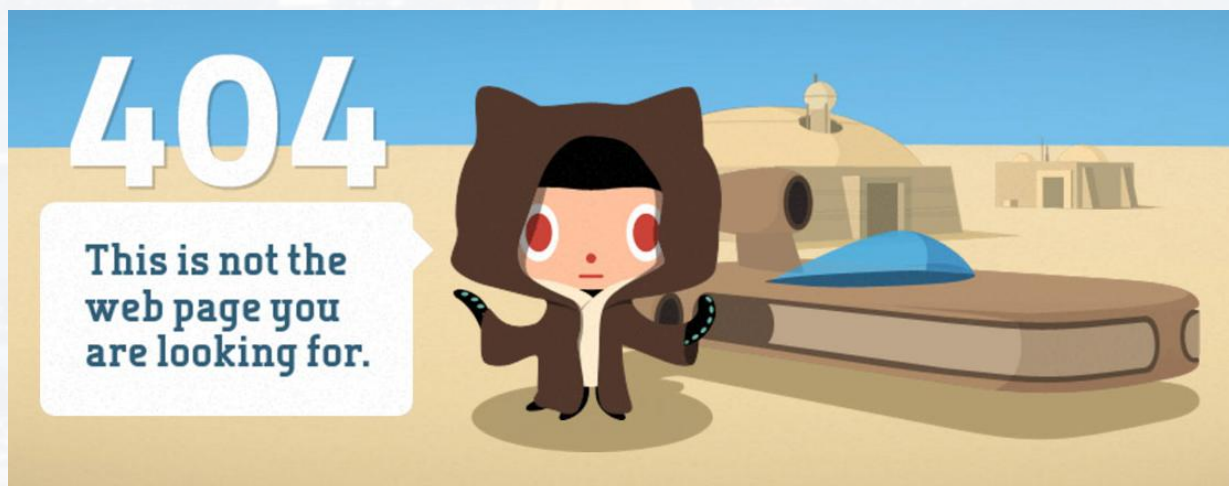
- **Git Hub**

create your repo

invite a co worker

push

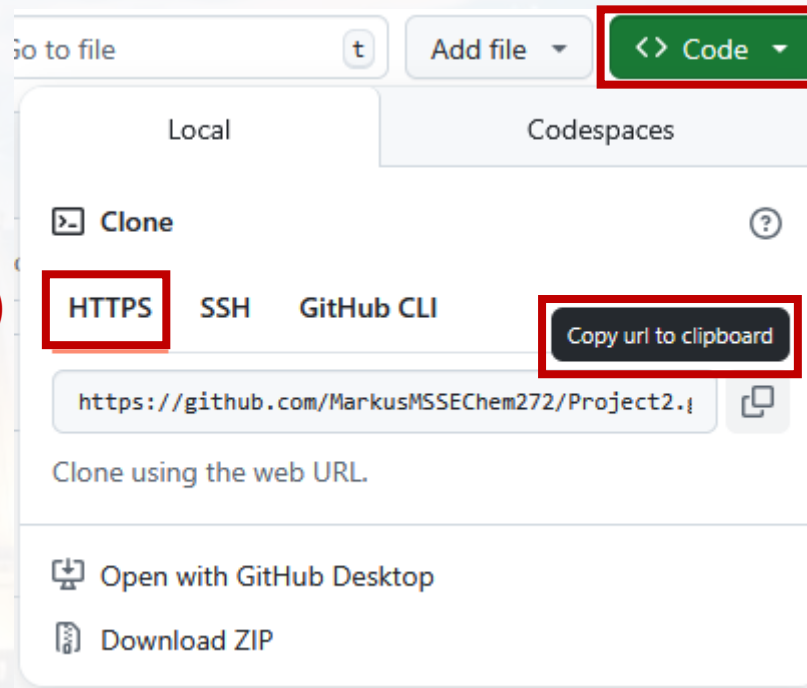
fetch, merge, pull





1) copying the remote repo to your local machine (`git clone`)

- copy the **url** of your repo



a) go to your repo and click *code*

b) select an option (e. g. HTTPS)

c) copy the url

next: open VS Code and navigate to the directory in which you want to copy the repo



next: open VS Code and navigate to the directory in which you want to copy the repo

```
:~/Berkeley/MSSE/Chem272/Projects/Project 2$
```

clone the repo:

```
git clone https://github.com/MarkusMSSEChem272/Project2.git
```

the url we copied earlier

```
Cloning into 'Project2'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
```

the entire repo is being
copied to your local
machine now!

```
● (base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem272/Projects/Project 2$ ls
OneHotEncoder.py Project2
```

```
● (base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem272/Projects/Project 2/Project2$ ls
README.md
```




2) workflow

a) navigate to your project folder

```
● (base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem272/Projects/Project 2/Project2$ ls  
README.md
```

b) check git status

either Git tells you that to local repo is up to date, or

it tells you that the **remote repo is X commits ahead**

→ run `git fetch` to update your local repo (**see later!**)!

c) you work on your code and want to add it to the remote repo now

```
git add .  
git commit -m "my message"
```

```
git push https://github.com/MarkusMSSEChem272/Project2.git main
```

pushes the new code
to the remote repo



2) workflow

```
git add .  
git commit -m "my message"
```

pushes the new code to the remote repo. You can replace the url with origin

```
git push https://github.com/MarkusMSSEChem272/Project2.git main
```

```
(base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem272/Projects/Project 2/Project2$ git push https://github.com/MarkusMSSEChem272/Project2.git main  
/mnt/c/Program Files/Git/mingw64/bin/git-credential-manager-core.exe get: 1: /mnt/c/Program Files/Git/mingw64/bin/git-credential-manager-core.exe: not found  
/mnt/c/Program Files/Git/mingw64/bin/git-credential-manager-core.exe store: 1: /mnt/c/Program Files/Git/mingw64/bin/git-credential-manager-core.exe: not found  
Enumerating objects: 4, done.  
Counting objects: 100% (4/4), done.  
Delta compression using up to 8 threads  
Compressing objects: 100% (3/3), done.  
Writing objects: 100% (3/3), 807 bytes | 807.00 KiB/s, done.  
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0  
To https://github.com/MarkusMSSEChem272/Project2.git  
97d088a..3902b95  main -> main
```

code is online!

Project2 Public

main 1 Branch 0 Tags

Go to file Add file Code

MarkusMSSEChem272 adding onehot enc 3902b95 · last week 3 Commits

OneHotEncoder.py	adding onehot enc	last week
README.md	Update README.md	last week

watch:

12i_GitHub_CloneRepo

12j_GitHub_Push



GitHub

Outline

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status, add and commit

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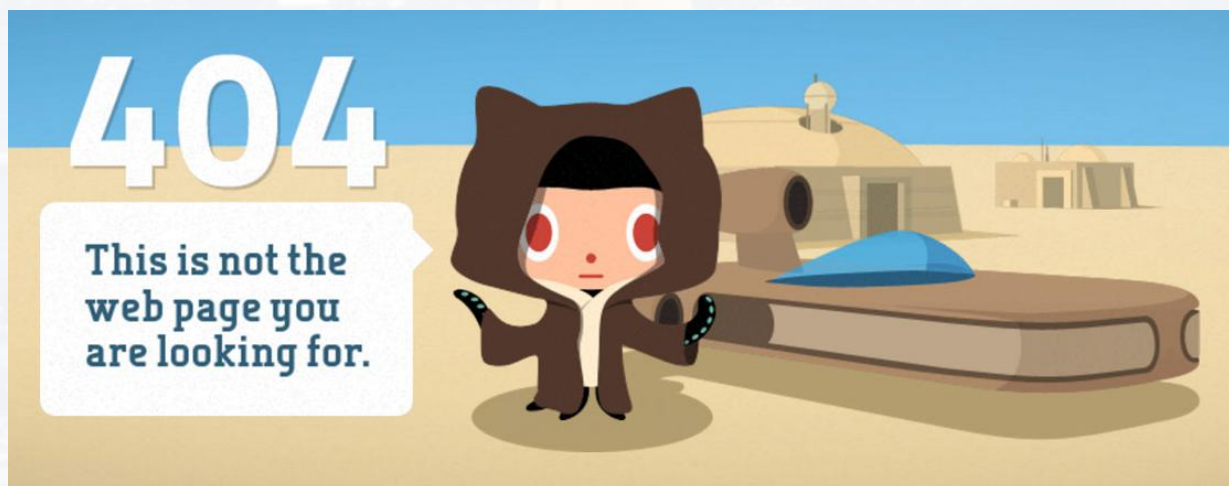
- **Git Hub**

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fetch, merge, pull

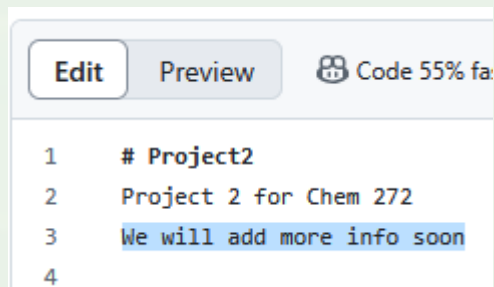




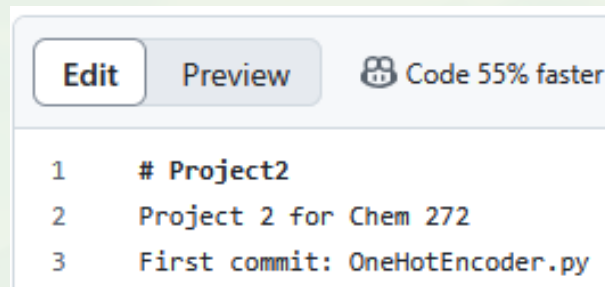
if your co-workers have pushed new code to the **remote repo**

→ update your **local repo/directory** accordingly

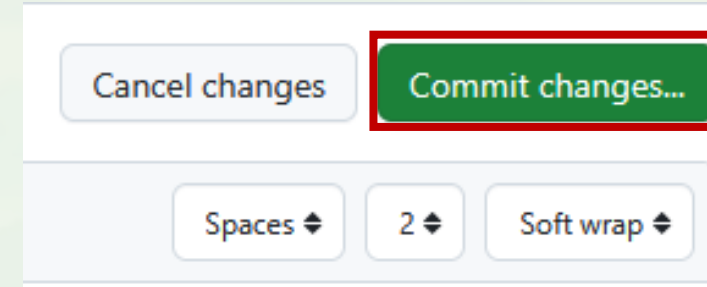
remote repo
(GitHub)



```
1 # Project2
2 Project 2 for Chem 272
3 We will add more info soon
4
```



```
1 # Project2
2 Project 2 for Chem 272
3 First commit: OneHotEncoder.py
```



Cancel changes **Commit changes...**

Spaces 2 Soft wrap

local repo

git status

```
● (base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem272/Projects/Project 2/Project2$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)
```




if your co-workers have pushed new code to the **remote repo**

→ update your **local repo/directory** accordingly

local repo

git status

```
● (base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem272/Projects/Project 2/Project2$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)
```

git fetch

```
● (base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem272/Projects/Project 2/Project2$ git fetch
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1000 bytes | 142.00 KiB/s, done.
From https://github.com/MarkusMSSEChem272/Project2
   efeafe4..d82475c  main       -> origin/main
```

git fetch

- **does not** update your working directory
- updates only commits, tags etc → **safe to inspect changes**

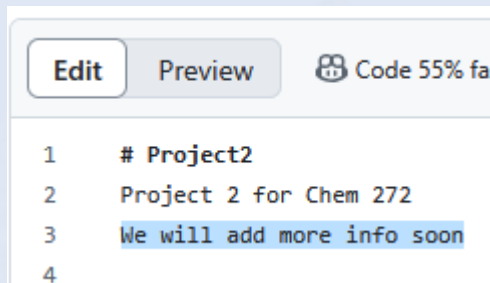


if your co-workers have pushed new code to the **remote repo**
→ update your **local repo/directory** accordingly

local repo

git fetch

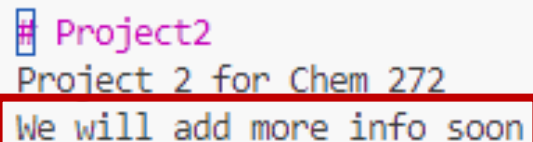
- **does not** update your working directory
- updates only commits, tags etc → **safe to inspect changes**



```
Edit Preview  Code 55% fa
1 # Project2
2 Project 2 for Chem 272
3 We will add more info soon
4
```

old version is still stored in working directory

working directory



```
# Project2
Project 2 for Chem 272
We will add more info soon
```

git merge

- **does** update your working directory

```
(base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem272/Projects/Project 2/Project2$ git merge
Updating 92d4c09..d82475c
Fast-forward
 README.md | 2 +--
 1 file changed, 1 insertion(+), 1 deletion(-)
```



if your co-workers have pushed new code to the **remote repo**
→ update your **local repo/directory** accordingly

**working
directory**

```
# Project2
Project 2 for Chem 272
We will add more info soon
```

git merge

- **does** update your working directory

```
(base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem272/Projects/Project 2/Project2$ git merge
Updating 92d4c09..d82475c
Fast-forward
 README.md | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)
```

```
# Project2
Project 2 for Chem 272
First commit: OneHotEncoder.py
```

Now, also the **working directory** has been updated

alternatively: git pull

runs git fetch **and**
git merge

```
(base) mmh_user@DESKTOP-PPSA666:~/Berkeley/MSSE/Chem272/Projects/Project 2/Project2$ git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 1011 bytes | 337.00 KiB/s, done.
From https://github.com/MarkusMSSEChem272/Project2
 d82475c..bce999b main -> origin/main
Updating d82475c..bce999b
Fast-forward
 README.md | 1 +
 1 file changed, 1 insertion(+)
```

watch:

12k_Git_Fetch_Merge_Pull



summary

`git add <my_file>`

adds a file to staging area

`git commit -m "my_message"`

commits files from staging area to local repo

`git push <remote_repo_name> <branch_name>`

updates remote repo

`git restore <my_file>`

undoes changes in indexed file

`git restore --staged <my_file>`

moves file from staged area to index

`git restore --source=HEAD~X`

moves commit (X commits before current one) back to staged area

`git fetch`

updates local commit from remote repo

`git merge`

updates local directory after fetch

`git pull`

runs fetch **and** merge

`git clone`

clones entire remote repo (has to be done only **once!**)



summary

```
git add <my_file>
```

```
git commit -m "my_message"
```

```
git push <remote_repo_name> <branch_name>
```

```
git restore <my_file>
```

```
git restore --staged <my_file>
```

```
git restore --source=HEAD~X
```

```
git fetch
```

```
git merge
```

```
git pull
```

```
git clone
```

adds a file to staging area

commits files from staging area
to local repo

updates remote repo

undoes changes in indexed file

moves file from staged area to index

moves commit (X commits before
current one) back to staged area

updates local commit from remote repo

updates local directory after fetch

runs fetch **and** merge

clones entire remote repo (has to be
done only **once**!)

note, there is way more you could do

check out different branches

moving head

We hope you have enjoyed the course!

